HLD-as-Source-of-Intent: Bootstrapping Intent-Based Networks from High-Level Design

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***Abstract*—Intent-Based Networking (IBN) is usually studied as an operational layer applied to an already deployed network []. This draft paper argues that the same intent-centric reasoning can be pushed earlier in the lifecycle, using a High-Level Design (HLD) document as the initial source of intent for bootstrapping a network from design to first deployment. We target small and medium enterprise networks and propose a closed-loop workflow that ingests an HLD, derives implementation artefacts, validates them in a lab or digital-twin environment, and produces a first operational network with human approval at critical steps []. The prototype relies on an agentic orchestration harness, a structured source of truth, and vendor-aware rendering, but these are implementation choices rather than the paper's primary claim. The main contribution is the scientific positioning of "HLD-as-intent" at time zero, its grounding in IBN concepts from RFC 9315 and RFC 9316 [], and an evaluation methodology across multiple HLD instances compatible with CNSM's focus on automation, intent, and agentic intelligence.**

***Index Terms*—intent-based networking, high-level design, network automation, closed-loop management, network design, digital twin, agentic systems**

I. Introduction

Intent-Based Networking (IBN) is most often introduced once a network already exists: operators define intents for a running infrastructure, and the system translates them into operational actions such as deployment, assurance, and remediation []. In practice, however, enterprise networks start earlier with human-written design artefacts: requirement documents, architecture decisions, and especially High-Level Designs (HLDs). This paper asks whether that HLD can become the first machine-actionable source of intent.

- **Problem statement.** There is a gap between the HLD, which captures architecture and rationale, and the deployed network, which is usually produced through manual, vendor-specific implementation steps.
- **Scientific angle.** Existing IBN work largely emphasizes operation on an existing network; we instead focus on the *T0* phase, where no network exists yet and the challenge is to bootstrap a first operational state from design-time intent [].
- **Working concept.** We treat this early-stage intent as a form of "Intent Zero": intent before operations, before assurance, and before the network has become a managed object.
- **Scope choice.** This paper deliberately limits itself to design-to-initial-deployment. A later operational IBN loop may coexist with this approach, but is not the primary object of study here.

The paper is built around the following claim: an HLD, provided in a structured and parsable form, can act as the initial source of intent for an end-to-end workflow that produces a first deployable network configuration, validates it in a controlled environment, and prepares a network for subsequent operation and management.

A. Draft Contributions

- **C1.** A reframing of IBN from *operations on an existing network* to *bootstrapping a network from an HLD*.
- **C2.** A lifecycle view that positions the HLD as the earliest explicit intent artefact, bridging design, dimensioning, implementation planning, and initial deployment.
- **C3.** A prototype closed-loop workflow, grounded in RFC terminology, that transforms HLD information into deployment artefacts, validation steps, and as-built feedback [].
- **C4.** An evaluation methodology based on multiple SME HLDs, vendor and constraint profiles, and human-in-the-loop checkpoints.

B. Paper Structure

- Section II provides background, motivation, and the positioning against related work.
- Section III presents the proposition at a high level.
- Section IV details the input model, workflow, and implementation choices.
- Section V defines the experimental evaluation plan.
- Section VI discusses scope boundaries, limitations, and implications.
- Section VII concludes.

II. Background and Motivation

A. Network Lifecycle and the Role of HLD

- Typical lifecycle: business needs $\rightarrow$ requirements $\rightarrow$ architecture $\rightarrow$ HLD $\rightarrow$ LLD $\rightarrow$ implementation planning $\rightarrow$ deployment $\rightarrow$ validation $\rightarrow$ operations.
- In current engineering practice, the HLD is often authoritative for humans but inert for machines.

- The HLD usually contains the architectural intent that matters most at T0: site structure, service zones, segmentation goals, interconnection principles, resilience expectations, and acceptance criteria.
- The translation from HLD to device-ready artefacts remains costly, manual, and error-prone in multi-vendor contexts [].

B. IBN Concepts Relevant to This Paper

- RFC 9315 defines IBN as a closed-loop system that converts high-level intent into network behavior and continuously checks conformance [].
- RFC 9315 also distinguishes an inner control loop and an outer human loop; this distinction is useful here, but our paper concentrates on the earliest fulfilment stages rather than steady-state operation [].
- RFC 9316 can be used to clarify what kind of intent is involved in this paper: business intent, service intent, resource intent, and the transformations between them [].
- One key question for the paper is therefore not only "what is intent?", but also "at what point in the lifecycle does intent become operationally actionable?".

C. Related Work and Research Gap

- **Classical IBN literature and platforms.** Most approaches assume an already deployed network and focus on policy translation, closed-loop assurance, or service steering [].
- **Recent CNSM/NOMS work.** Existing work covers intent translation, traffic steering, NFV or Kubernetes deployment, and KPI assurance with LLMs, but does not clearly address the HLD-to-network bootstrap problem [].
- **MALT-inspired abstraction.** Multi-layer modelling work such as MALT helps justify the need for several abstraction levels across a network lifecycle, but does not provide an open demonstrator for turning HLD content into an initial running network [].
- **Gap.** We did not identify a clear scientific treatment of HLD-as-source-of-intent for initial network realization from a white-field starting point.
- **Positioning.** Our paper is therefore closer to *design-time IBN* or *Intent Zero* than to classical operational IBN.

III. Proposition: Overview

A. Core Proposition

- **Input.** A structured HLD describing the target SME network, possibly complemented by explicit technology constraints.
- **Output.** A first validated network realization: topology artefacts, vendor-aware configurations, deployment plan, and as-built feedback.
- **Central hypothesis.** A sufficiently structured HLD can serve as the first formalized intent artefact without requiring an operator to re-enter the same intent into a separate controller-specific model.
- **Boundary.** The paper does not claim to solve the whole lifecycle; it addresses the transition from design intent to first deployable and validated network state.

B. Closed-Loop View

- Step 1: ingest and parse the HLD into structured design objects.
- Step 2: normalize those objects into an internal intent representation.
- Step 3: derive policy, topology, addressing, segmentation, and implementation artefacts.
- Step 4: validate the result in a lab or digital-twin environment before production push [].
- Step 5: generate deployment and validation outputs for human review when changes are high impact.
- Step 6: feed the observed result back into an as-built view that can later support operations.

C. Why a Closed Loop at T0?

- Because design artefacts and implementation artefacts drift even before the network enters normal operation.
- Because a design-time loop can catch mismatches early: missing constraints, unsupported vendor features, invalid topology assumptions, and inconsistent security intent.
- Because a digital twin or lab validation stage can reduce the risk of turning an HLD directly into production changes [].
- Because this first loop can later hand over a cleaner, better documented network to a classical operational IBN loop.

IV. Details

A. Input Model: HLD or Structured Input File

- Current working assumption: the entry artefact is still called an HLD, but it must contain structured sections that are parsable and reproducible.
- Candidate sections: scope, sites, zones, topology intent, addressing, segmentation, external connectivity, security rules, critical services, vendor or platform constraints, validation expectations.
- Open design choice to discuss in the paper: whether technology constraints belong inside the HLD or in a companion constraint file.
- Candidate formalizations to compare in the future: Markdown with tables, PlantUML, Mermaid, or another machine-readable design notation [].

B. Transformation Workflow

- **Design parsing.** Extract structured entities and relations from the HLD.
- **Intent normalization.** Convert raw HLD content into a stable internal representation that distinguishes invariant design choices from platform-dependent constraints.
- **Policy and topology derivation.** Produce network intents such as segmentation, reachability, redundancy, and service exposure.
- **Implementation synthesis.** Render vendor-aware artefacts, low-level configurations, topology descriptors, and deployment plans.
- **Validation.** Deploy in Containerlab or an equivalent environment, execute validation checks, and collect discrepancies.

- **As-built feedback.** Record the realized state and discrepancies to support traceability and later operation.

C. Prototype Realization Choices

- The current prototype uses an agentic orchestration harness to decompose the workflow into specialized tasks.
- The present draft intentionally treats this multi-agent decomposition as a *means of realization*, not as the paper's main scientific claim.
- A graph-backed source of truth helps preserve cross-layer relations between design objects, implementation artefacts, deployment state, and validation results [].
- Additional memory layers can be useful for rationale capture, traceability, and human interaction, but should remain secondary in the paper narrative.
- Vendor-specific rendering and deployment backends are required to demonstrate that the approach is not limited to a single platform [].

D. Human-in-the-Loop and Governance

- The workflow should remain largely automated for low-risk transformations.
- A human checkpoint is required for high-impact changes: new devices, topology modifications, security-sensitive exposure, or unresolved implementation ambiguity.
- One evaluation dimension is therefore not only technical correctness, but also the *amount and type of human intervention* still required.
- This is also where closed-vendor documentation, unsupported models, or ambiguous HLD wording become visible.

V. Experimental Evaluation

A. Objectives and Research Questions

- **RQ1.** Can a structured HLD provide enough information to generate a first deployable SME network without manual re-entry into another intent model?
- **RQ2.** How robust is the workflow across multiple HLDs, topology shapes, and vendor or technology constraints?
- **RQ3.** Where is human intervention still required, and is that intervention due to HLD ambiguity, missing constraints, or implementation limitations?
- **RQ4.** To what extent are the results dependent on a specific model, toolchain, or vendor backend?

B. Experimental Setup

- Target domain: small and medium enterprise networks.
- Validation environment: Containerlab or another comparable digital-twin infrastructure [].
- Multiple vendor profiles: for example software routers, Linux-based network functions, and model-driven platforms.
- Reproducible pipeline execution from the same HLD input to the same expected artefacts and validation checks.

C. Scenario Corpus

- A corpus of roughly ten HLDs with increasing complexity: single site, multi-site, DMZ exposure, guest access, service segmentation, redundancy, and vendor heterogeneity.
- Mix of synthetic and, when available, industrially inspired HLDs.
- Optional stress cases: underspecified HLD, conflicting constraints, unsupported vendor feature, and closed-vendor documentation dependency.

D. Metrics

- End-to-end success rate per HLD.
- Number of manual interventions per scenario and their causes.
- Time to first validated network realization.
- Coverage of HLD elements successfully translated into deployment artefacts.
- Validation pass rate: connectivity, segmentation, reachability, and service tests.
- Reproducibility across repeated runs.
- Optional cost indicators: tokens, latency, and model dependence when an LLM-assisted component is enabled.

E. Comparative and Ablation Angles

- Compare structured-HLD input with less structured variants to quantify the value of formalization.
- Compare one combined input file against a split HLD-plus-constraints organization.
- Compare validation-only digital-twin execution against a path that targets a more realistic deployment backend.
- Ablate optional components such as memory support or different orchestration strategies to show what is essential to the core claim.

VI. Discussion

A. Scientific Positioning

- The paper should explicitly state that it is *not* a full replacement for operational IBN.
- Instead, it introduces a preceding phase: design-time intent realization, or "Intent Zero", which prepares a network before classical assurance and operations begin.
- This clarifies the apparent tension raised by the draft title: the system is IBN-inspired and RFC-grounded, but it operates earlier in the lifecycle than most existing IBN work.

B. Limits and Threats to Validity

- Results may depend on the structure and quality of the HLD itself.
- A lab or digital twin remains a proxy for production infrastructure, even if it is useful for pre-deployment validation.
- Vendor closedness, missing YANG or API models, and incomplete documentation may require extra human intervention.
- The chosen orchestration architecture may influence engineering efficiency without changing the main scientific claim.

C. Implications

- For IBN research: intent should be studied not only as an operational abstraction, but also as a design-time artefact.
- For practitioners: a disciplined HLD format may reduce the cost of moving from architecture to implementation.
- For future work: the same framework could later be extended toward an operational IBN loop, continuous assurance, and design updates after deployment.

VII. Conclusion

- This paper draft positions the HLD as the earliest explicit source of network intent.
- The main message is that the gap between design and deployment can be framed as an IBN problem, not only as a documentation or automation problem.
- The proposed workflow aims to transform a structured HLD into a first validated network realization for SME environments.
- The next steps are to finalize the HLD formalization choice, complete the experimental campaign across multiple HLDs, and sharpen the discussion on coexistence with later operational IBN loops.

References

References will be added after the current `\cite{}` placeholders are replaced with actual keys.